Lab 2# - Breathing Strap

BME/CS 479 - Wearables and Nearables Technology Laboratory – Spring 2026 Rafael Mejia | Group 7

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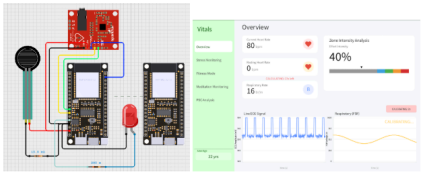
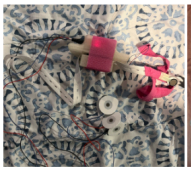
Hardware Design and Development

In this project we used the FireBeetle 328P which was the brain, transmitting data via BLE to Processing UI. SparkFun AD8232 Heart Rate Monitor which has leads on the skin and sends metric data to the Arduino. The FSR was attached to the strap to collect inhaling and exhaling data. We designed the first wearable so that the user can wear it on their chest. The second wearable was made more stylish as a vest. Some assembly challenges were the 3D printing as it would not fully print or it would break on some attempts. There were also some issues with the sensors not working and the pads losing stickiness. We had to find the sweet spot for these pads. To filter data we made sure to send medians to the UI.



Breathing Strap:



Vest Attachment:

Figure: The image shows the assembled device (2 inches wide × 2 inches high) and also includes the circuit diagram and UI screenshot.

User Interface Design and Development

The user interface was designed for simplicity and to “get to the point”. The home screen of the user interface has 4 options where each of these are the requirements for the lab plus a bonus PSC analysis tab. The graph also shows colors of each zone so that they do not need to look at numbers just colors to see where they are. The axis also has a combination of scrolling and static so that they can see previous data but also see new data. On the Arduino we do the calculations to reduce the load and dependency of it from the UI. We send metrics in medians to keep it consistent. One thing we can improve on the code is calibrating on the arduino automatically and not on the user interface manually.